

Akura: Dyslexia Aware Automated Sinhala Essay Grading System

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Abstract—Automated Essay Grading systems offer significant benefits in reducing teacher workload and enabling scalable assessment, yet most existing approaches fail to adequately support students with learning differences such as dyslexia. This limitation is particularly evident in low-resource languages like Sinhala, where the abugida script and complex diacritic system give rise to distinctive dyslexic writing patterns that are poorly handled by conventional grading models. Consequently, Sinhala-speaking dyslexic students are often unfairly penalized for mechanical and orthographic errors rather than being evaluated on their underlying conceptual understanding. This research proposes Akura, a dyslexia-aware automated Sinhala essay grading system designed to support fair, inclusive, and cognitively informed assessment. The system integrates four complementary components: a Sinhala dyslexic writing pattern classifier for identifying language-specific cognitive error types, a context-aware sentence reconstruction module that restores intended meaning from distorted text, a bias-aware scoring engine that evaluates content while mitigating discrimination against dyslexic writing patterns, and a multimodal feedback platform that delivers accessible visual and auditory feedback to learners and educators. By combining Sinhala-specific natural language processing, human-in-the-loop validation, and fairness-aware machine learning techniques, Akura aims to bridge the assessment gap between dyslexic and non-dyslexic learners. The proposed system contributes toward equitable educational assessment in Sri Lanka while advancing research on fairness-driven automated essay grading for low-resource and neurodiverse language contexts.

Keywords—Dyslexia, Automated Essay Grading, Sinhala Language, Fairness Aware Machine Learning, Natural Language Processing, Educational Technology

I. INTRODUCTION (HEADING I)

This template, modified in MS Word 2007 and saved as a “Word 97-2003 Document” for the PC, provides authors with most of the formatting specifications needed for preparing electronic versions of their papers. All standard paper components have been specified for three reasons: (1) ease of use when formatting individual papers, (2) automatic compliance to electronic requirements that facilitate the concurrent or later production of electronic products, and (3) conformity of style throughout a conference proceedings. Margins, column widths, line spacing, and type styles are built-in; examples of the type styles are provided throughout this document and are identified in italic type, within parentheses, following the example. Some components, such as multi-leveled equations, graphics, and tables are not prescribed, although the various table text styles are provided. The formatter will need to create these components, incorporating the applicable criteria that follow.

II. LITERATURE REVIEW

Automated Essay Scoring (AES) systems have been widely adopted to reduce teacher workload and enable scalable educational assessment. Early AES models focused on surface-level linguistic and statistical features and achieved high agreement with human raters. However, recent studies highlight that such systems may unintentionally inherit and amplify human rater bias, leading to unfair scoring outcomes across learner populations.

A. Bias in Automated Essay Scoring

Amorim *et al.* demonstrated that AES models trained on biased human ratings tend to reproduce subjective rater behavior rather than objectively assessing writing quality[1]. Their findings show that biased rater comments introduce

distortions during model training, causing certain writing styles to be systematically penalized. Removing biased samples improved both fairness and agreement, indicating that bias in AES is a learned systemic issue rather than a simple data imbalance.

Guo conducted one of the earliest large-scale fairness evaluations of AES in high-stakes assessment using GMAT AWA data. Through subgroup analysis and propensity score matching, the study emphasized the necessity of empirical fairness evaluation when AES systems are deployed in consequential educational settings[2].

B. Fairness Evaluation and Bias Mitigation Techniques

General-purpose fairness frameworks have been introduced to address bias in machine learning systems. Bellamy *et al.* proposed AI Fairness 360 (AIF360), an open-source toolkit offering standardized fairness metrics such as Statistical Parity Difference, Disparate Impact Ratio, and Equal Opportunity Difference, along with bias mitigation techniques across multiple stages of the ML pipeline. While effective in general AI applications, these frameworks are not tailored to educational NLP or low-resource languages[3].

Counterfactual fairness approaches further extend fairness evaluation by examining whether predictions change under hypothetical modifications of sensitive attributes. Such methods reveal biases that group-level metrics may overlook, making them relevant for educational datasets where learner characteristics implicitly influence model behavior[4].

C. AES Datasets and Attribute-Level Scoring

Most AES datasets provide only holistic essay scores, limiting transparency and fairness analysis. Mathias and Bhattacharyya introduced ASAP++, enriching the widely used ASAP dataset with attribute-level scores such as content, organization, and fluency. This enables more interpretable and fine-grained scoring and supports fairness-aware evaluation by isolating the contribution of individual writing attributes[5].

D. AES in Low-Resource and Neurodiverse Contexts

While AES research has matured for English, low-resource languages remain underrepresented. Singh *et al.* demonstrated the feasibility of AES for Hindi using transformer-based models but highlighted challenges related to data scarcity and inconsistent scoring behavior[6]. Similar limitations are observed in Nepali AES studies, reinforcing the structural disadvantages faced by low-resource languages.

In the Sinhala context, de Silva provided a comprehensive survey of publicly available NLP tools and resources, identifying significant gaps in standardized datasets and educational NLP applications[7]. Existing Sinhala tools primarily focus on surface-level spelling and grammar correction, which risks misclassifying dyslexia-related variations as generic errors.

Research on neurodiversity further indicates that surface-level features such as spelling and punctuation

disproportionately bias writing quality assessments for dyslexic learners. These findings emphasize the need for assessment systems that prioritize semantic meaning over orthographic correctness[1].

E. Research Gap

Existing research provides strong foundations in AES, fairness-aware machine learning, and low-resource NLP. However, no prior work integrates semantic-aware AES, fairness metrics, and neurodiversity adaptation within the Sinhala educational context. Current Sinhala AES systems lack fairness evaluation, while fairness toolkits are not adapted for low-resource languages or dyslexic writing patterns. This gap motivates the development of a bias-aware Sinhala essay grading system that ensures equitable assessment for both dyslexic and non-dyslexic learners[1].[2][3]

F. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m²” or “webers per square meter”, not “webers/m²”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm³”, not “cc”. (*bullet list*)

G. Equations

The equations are an exception to the prescribed specifications of this template. You will need to determine whether or not your equation should be typed using either the Times New Roman or the Symbol font (please no other font). To create multileveled equations, it may be necessary to treat the equation as a graphic and insert it into the text after your paper is styled.

Number equations consecutively. Equation numbers, within parentheses, are to position flush right, as in (1), using a right tab stop. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Note that the equation is centered using a center tab stop. Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

H. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

III. USING THE TEMPLATE

After the text edit has been completed, the paper is ready for the template. Duplicate the template file by using the Save As command, and use the naming convention prescribed by your conference for the name of your paper. In this newly created file, highlight all of the contents and import your prepared text file. You are now ready to style your paper; use the scroll down window on the left of the MS Word Formatting toolbar.

A. Authors and Affiliations

The template is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

1) *For papers with more than six authors:* Add author names horizontally, moving to a third row if needed for more than 8 authors.

2) *For papers with less than six authors:* To change the default, adjust the template as follows.

a) *Selection:* Highlight all author and affiliation lines.

b) *Change number of columns:* Select the Columns icon from the MS Word Standard toolbar and then select the correct number of columns from the selection palette.

c) *Deletion:* Delete the author and affiliation lines for the extra authors.

B. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced. Styles named “Heading 1”, “Heading 2”, “Heading 3”, and “Heading 4” are prescribed.

C. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE I. TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
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^a Sample of a Table footnote. (Table footnote)

Fig. 1. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT (*Heading 5*)

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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To have non-visible rules on your frame, use the MSWord “Format” pull-down menu, select Text Box > Colors and Lines to choose No Fill and No Line.